

Knowledge Map

What I Know, What's Known, and What's Open

Patrick S. Mahon

Personal reference document — not for submission

May 2026

Abstract

A consolidated reference covering everything explored in this research programme: what I understand, what is known (and by whom), what I proved that turned out to be classical, what remains genuinely open, and where the understanding might lead. Not a paper. A map for finding the next problem.

Contents

1	The central question	2
2	Layer 1: From charges to measures	2
2.1	The classical results (all known)	2
2.2	What I proved (turned out classical)	2
2.3	What I understand from this	3
3	Layer 2: From dual space to realization	3
3.1	The classical results (all known)	3
3.2	What I proved (turned out classical)	3
3.3	What I understand from this	3
4	Layer 3: From probability to value-definiteness	4
4.1	The classical results (all known)	4
4.2	What I proved (classical + one novel framing)	4
4.3	What I understand from this	4
5	Delay reconstruction	5
5.1	The classical results	5
5.2	What I proved (mostly classical)	5
5.3	What is genuinely novel (but motivation unclear)	5
5.4	What I understand from this	6
6	Dead ends and lessons	6
7	What remains genuinely open	6
8	The understanding, condensed	7
9	Key references	7

1 The central question

When does observationally coherent data determine a genuine probability measure, and what additional commitments does this require?

This decomposes into three layers:

1. **Extension:** compatible finitely additive charges on a directed system of Boolean algebras $\rightarrow$ σ -additive probability measure.
2. **Realization:** measure on the dual space (Stone space) $\rightarrow$ measure on an underlying state space Ω .
3. **Value-definiteness:** probabilistic description $\rightarrow$ simultaneous definite values for all observables.

Each layer requires a commitment the previous does not. Distributivity of the observation algebra controls which commitments are available.

2 Layer 1: From charges to measures

2.1 The classical results (all known)

- **Continuity from above:** A finitely additive charge on an algebra is σ -additive iff $E_n \downarrow \emptyset \Rightarrow \ell(E_n) \rightarrow 0$. *Halmos, Measure Theory (1950), §13, Theorem D.*
- **Carathéodory extension:** A σ -additive premeasure on a semiring extends uniquely to a measure on the generated σ -algebra. *Carathéodory (1914); Halmos §13; Bogachev Vol. 1, Ch. 1.*
- **Kolmogorov extension:** Compatible finite-dimensional distributions on a product space extend to a unique probability measure, under standard Borel / tightness conditions. *Kolmogorov, Grundbegriffe (1933); Billingsley, Probability and Measure, Ch. 36.*
- **Choksi inverse limits:** Compatible measures on an inverse system of compact Hausdorff spaces extend to the inverse limit. *Choksi, Inverse limits of measure spaces, Proc. London Math. Soc. (1958), Theorem 1.*
- **Yosida–Hewitt decomposition:** Every bounded finitely additive charge decomposes uniquely as $\ell = \ell_c + \ell_p$ where ℓ_c is σ -additive and ℓ_p is purely finitely additive. *Yosida & Hewitt, Finitely additive measures, Trans. AMS (1952).*

2.2 What I proved (turned out classical)

Collective exhaustion (CE) in a directed system: defined as $\exists j \geq i$ such that $\ell_j(\pi_{ij}^{-1}(E_n)) \rightarrow 0$. Under compatibility, the existential witness is vacuous: $\ell_j(\pi_{ij}^{-1}(E_n)) = \ell_i(E_n)$ for all $j \geq i$. So CE = σ -additivity of each ℓ_i . This is not a theorem; it is the observation that the definition contains a redundant quantifier.

The directed-system extension theorem (Theorem ?? in Paper I) is Kolmogorov extension in directed-system language with “sequential common refinements” replacing the usual product structure.

The Lean formalization (0 sorry on the Carathéodory route, 1 sorry for Yosida–Hewitt on the Stone route) is genuine verified infrastructure, even though the theorems are classical.

2.3 What I understand from this

1. The passage from finite additivity to σ -additivity is the *only* non-finitary step in probability. Everything else (compatibility, normalization, finite additivity) is first-order / finitary.
2. No first-order condition can force σ -additivity. *Immediate from Loś's theorem: the ultraproduct of Dirac masses on $\mathbb{N}$ is a purely finitely additive charge.* Hoover (1978), Fagin–Halpern–Megiddo (1990) for the probability-logic formulation.
3. The finite-cofinite charge on $\mathbb{N}$ is the canonical obstruction: normalized, finitely additive, not σ -additive, and arises as the ultraproduct witness.
4. The Yosida–Hewitt decomposition gives the exact characterization: $\text{CE} \Leftrightarrow \ell_p = 0$.

3 Layer 2: From dual space to realization

3.1 The classical results (all known)

- **Stone representation:** Every Boolean algebra is isomorphic to the clopen algebra of a compact, Hausdorff, totally disconnected space (its Stone space). *Stone (1936); Johnstone, Stone Spaces (1982).*
- **Charge $\leftrightarrow$ measure on Stone space:** Every finitely additive charge on a Boolean algebra B corresponds to a regular Borel (Baire) measure on $\text{St}(B)$. *Fremlin, Measure Theory, Vol. 3, §326 and Vol. 4, §416; Halmos (1950).*
- **σ -additivity $\leftrightarrow$ support:** A charge on B is σ -additive iff the corresponding Stone-space measure vanishes on meagre Baire sets, equivalently assigns full mass to countably complete ultrafilters. *Rao & Rao, Theory of Charges (1983), Theorem 10.5.3.*
- **Principal ultrafilters:** Every principal ultrafilter is countably complete. The converse holds for $\mathcal{P}(\kappa)$ when κ is below the first measurable cardinal (Ulam's theorem). For non- σ -complete algebras (typical cylinder algebras), countably-complete non-principal ultrafilters can exist without large cardinals.

3.2 What I proved (turned out classical)

Stone measure exists unconditionally: Any compatible charges produce a Baire probability measure $\hat{\mu}$ on $\text{St}(\mathcal{C})$. This is Stone representation + Choksi. Lean-verified (0 sorry).

Descent requires $\hat{\mu}(\text{pure}(\Omega)) = 1$: concentration on principal ultrafilters. Under discriminability, this is equivalent to σ -additivity. This is Rao–Rao 10.5.3 restated.

Realization is unconstrained: Any surjective-projecting $S \subseteq \text{St}(\mathcal{C})$ can serve as Ω . This is the abstractness of Boolean algebras recognized since Stone (1936).

3.3 What I understand from this

1. The Stone construction gives the *unconditional* object: $\hat{\mu}$ always exists. It is the “empirically adequate” object in van Fraassen’s sense.
2. Descent to Ω is a *choice*: which ultrafilters count as “realized” is additional structure over the algebra.

3. The three-layer nesting:

$$\text{pure}(\Omega) \subseteq \{\text{countably complete ultrafilters}\} \subseteq \text{St}(\mathcal{C}).$$

σ -additivity gives mass on the middle set (intrinsic). Descent needs mass on the inner set (depends on Ω).

4. The van Fraassen identification (Stone measure = EA, descent = realist commitment) has no precedent in the literature. It is a philosophical contribution, not a mathematical one.

4 Layer 3: From probability to value-definiteness

4.1 The classical results (all known)

- **Boolean algebras admit 2-valued homomorphisms:** Ultrafilters on Boolean algebras are exactly the 2-valued homomorphisms. Existence by Zorn. *Stone (1936); every Boolean algebra textbook.*
- **Kochen–Specker:** For $L(H)$, $\dim H \geq 3$, no dispersion-free state (2-valued homomorphism) exists. *Kochen & Specker (1967).*
- **Gleason:** For $L(H)$, $\dim H \geq 3$, every σ -additive state is $s(P) = \text{tr}(\rho P)$. *Gleason (1957).*
- **Bub–Clifton:** Given a state ρ and a preferred observable R , there is a unique maximal Boolean subalgebra $D(\rho, R) \subseteq L(H)$ on which the state is a mixture of dispersion-free states. *Bub & Clifton (1996); Halvorson & Clifton (1999).*
- **OML duality:** The category of orthomodular lattices is dually equivalent to a category of orthomodular spaces. Principal filters $P(A)$ are part of the dual structure (unlike Stone duality where they are invisible). *McDonald & Bimbó (2023), MLQ. Zero citations in quantum foundations as of 2026.*
- **Colimit generation:** Gluing Boolean algebras via categorical pushout produces orthomodular lattices, generically non-distributive. *Gunji, Haruna & Uragami (2026), arXiv:2603.22353.*

4.2 What I proved (classical + one novel framing)

Commensurability theorem: Distributivity is sufficient for VDR; for $L(H)$, $\dim H \geq 3$, non-distributivity blocks it. Lean-verified (0 sorry, 1 axiom for KS). *The mathematics is the assembly of Stone + KS + Gleason + Bub–Clifton. Not a new theorem.*

The EA/PR/VDR vocabulary connecting van Fraassen’s empirical adequacy to quantum logic via McDonald–Bimbó duality is novel framing. No precedent found.

The coherence filtration $\mathcal{S}(A) \supseteq \mathcal{S}_\sigma(A) \supseteq \mathcal{S}_{\text{df}}(A)$ with three named transitions (pasting, regularity, sharpness) organizes known results. The chain itself is textbook (Pták–Pulmannová, Kalmbach); the transition names and the observation that distributivity collapses the filtration are new packaging.

4.3 What I understand from this

1. **Distributivity controls value-definiteness.** Boolean algebras have ultrafilters (VDR available). Non-Boolean OMLs ($\dim \geq 3$) do not (KS). The dividing line is distributivity — sufficient but not necessary in full generality ($\dim 2$, horizontal sums).

2. **The algebra can constrain realization in the OML case.** Unlike Boolean Stone duality, the McDonald–Bimbó dual carries $P(A)$ as structure. “Realization is additional structure” in the Boolean case; it is *constrained* structure in the OML case.
3. **The realism debate has different mathematical answers for different algebras.** Boolean: EA, PR, VDR all available; philosophical choice. OML ($\dim \geq 3$): VDR blocked; PR available (Gleason); EA always available.
4. **The topos programme is more powerful.** Isham–Butterfield / Döring–Isham / Heunen–Landsman–Spitters encode the same structure via spectral presheaves and Bohri-fication. Our approach is coarser (single dual space, not presheaf over contexts) but connects more directly to van Fraassen.

5 Delay reconstruction

5.1 The classical results

- **Takens (1981):** Generic smooth observable gives an injective delay map for $L \geq 2 \dim X$.
- **Sauer–Yorke–Casdagli (1991):** Prevalence-based embedding theorem.
- **Generating partitions (Sinai 1959, Krieger 1970):** $h(T, P) = h(T)$ iff P generates.
- **Grassberger–Procaccia:** Correlation integral estimates D_2 and K_2 (Rényi-2 dimension and entropy) from delay data.
- **FNN (Kennel–Brown–Abarbanel 1992):** Embedding dimension selection. Sensitive to noise (Rhodes–Morari 1997).
- **Ragwitz–Kantz (2002):** Local prediction error minimization for (d, L) selection.
- **Botvinick-Greenhouse et al. (2025):** Measure-theoretic Takens via optimal transport. Lifts embedding to probability spaces.

5.2 What I proved (mostly classical)

Reconstruction theorem: Observable σ -algebra = full σ -algebra mod μ iff delay algebra dense in L^2 iff delay map is measure-theoretic embedding. *This is Krieger’s generating partition theorem in delay-map language.*

Predictive factorization, Chapman–Kolmogorov, Koopman–Perron: All classical (Rokhlin, Kolmogorov, Koopman 1931).

Finite-sample convergence rates: Minimax rates $n^{-s/(2s+d)}$ via Stone (1982); elbow stopping via Lepski (1991). *Not cited in the original work — a significant gap.*

5.3 What is genuinely novel (but motivation unclear)

Fibre mixing: A spatial non-degeneracy condition on the Rokhlin disintegration over delay fibres. Not equivalent to any standard temporal mixing condition. Closest relative: Furstenberg relative weak mixing (different condition). *Novel definition. Open derivability question.*

Bridge theorem: Algebraic reconstruction ($\delta \rightarrow 0$) $\Leftrightarrow$ geometric reconstruction (collision probability $\rightarrow 0$) under fibre mixing. *Novel. But: on standard Borel spaces with fixed L , the two conditions are essentially equivalent anyway. The theorem is about convergence rates as L grows. Motivation unclear — needs an example where the two modes actually diverge.*

Entropy characterization: $\delta(L) \rightarrow 0$ iff $H_2(\nu_L) \rightarrow \infty$. *Novel. But: a reinterpretation of what Grassberger–Procaccia already computes, not a new tool.*

5.4 What I understand from this

1. The reconstruction problem is *algebraically underdetermined*: the observation algebra does not constrain the realization. Embedding selection (d, L) is necessarily empirical.
2. Every practical reconstruction criterion (FNN, MI, Lyapunov convergence, prediction error) implicitly assumes a modelling commitment. Deterministic criteria fail under noise because noise violates the commitment.
3. The Rokhlin distance $\delta(\mathcal{O})$ is the measure-theoretic notion of “how close is the observable σ -algebra to generating everything.” It is the Rokhlin metric, named and studied since 1967.
4. Collision probability / Rényi-2 entropy measures the “collision structure” of the fibre decomposition. Grassberger–Procaccia already extracts this from data.
5. Fibre mixing is about the *transition regime*: fibres shrinking but not yet singletons. It is vacuous at both endpoints (no reconstruction, and full reconstruction).

6 Dead ends and lessons

1. **De Finetti via Stone:** Exchangeability forces σ -additivity (Hausdorff moments). The “exotic part” of the Stone measure is empty. Dead end.
2. **Stationary Stone:** The Yosida–Hewitt decomposition commutes with shift-invariance. An observation, not a theorem. The real open question (ergodic theory of the purely finitely additive part) is out of reach.
3. **Dynamics from observation:** “Can the Stone space give rise to dynamics without positing T ?” No: the algebra doesn’t constrain Ω , so it certainly doesn’t constrain structure on Ω .
4. **Paper III (noise thresholds):** Residual autocorrelation as embedding diagnostic = Billings–Voon (1986) / Casdagli (1991). FNN noise sensitivity = Rhodes–Morari (1997). Elbow stopping = Lepski (1991). Rediscovery throughout.
5. **“No intermediate condition”:** The claim that no condition interpolates between first-order and σ -additivity is false. $\limsup \ell(A_n) < c$ trivially interpolates.

7 What remains genuinely open

1. **Fibre mixing derivability:** Does ergodicity force ν -a.e. large fibre to be balanced for near-maximizers of the Rokhlin distance? *Well-posed, partially solved (positive fraction proved), but the paper built around it may lack motivation. Need a motivating example first.*
2. **OML extension problem:** Does every state on a general OML extend to a σ -additive measure on the McDonald–Bimbó dual $S_0(A)$? Gleason answers for $L(H)$; the general case is open. *Needs OML expertise to pursue.*
3. **Ergodic theory of purely finitely additive charges:** What are the extreme points of the purely finitely additive shift-invariant charges? Connected to Banach limits. *Out of reach currently.*

8 The understanding, condensed

One sentence per insight:

1. The passage from coherent charges to probability requires σ -additivity, which is not derivable from finitary conditions.
2. The Stone construction gives an unconditional measure on the dual space; descent to a realization requires a commitment the algebra cannot force.
3. Which ultrafilters are “realized” is additional structure in the Boolean case and constrained structure in the OML case.
4. Distributivity is sufficient for value-definiteness; non-distributivity ($\dim \geq 3$) blocks it.
5. The realism/empiricism boundary is algebraic: van Fraassen’s empirical adequacy is the unconditional level; value-definite realism requires distributivity.
6. Delay reconstruction is algebraically underdetermined: embedding selection is necessarily empirical.
7. Fibre mixing is a novel spatial condition on the disintegration, distinct from temporal mixing, relevant only in the transition regime.
8. The bridge between algebraic and geometric reconstruction may be solving a problem that doesn’t arise naturally.
9. Every practical reconstruction diagnostic implicitly assumes a modelling commitment; noise violates deterministic commitments.
10. Audit before you draft.

9 Key references

Measure extension and charges

- Halmos, *Measure Theory* (1950) — §13, §53–54
- Billingsley, *Probability and Measure* — Ch. 36
- Bogachev, *Measure Theory* — Vol. 1, Ch. 1
- Rao & Rao, *Theory of Charges* (1983) — Thm 10.5.3
- Yosida & Hewitt (1952) — Trans. AMS
- Choksi (1958) — Proc. London Math. Soc.
- Fremlin, *Measure Theory* — Vol. 3 §326, Vol. 4 §416

Stone duality

- Johnstone, *Stone Spaces* (1982)
- Stone (1936) — Trans. AMS

Quantum logic and contextuality

- Kalmbach, *Orthomodular Lattices* (1983)
- Pták & Pulmannová, *Orthomodular Structures as Quantum Logics* (1991)
- Kochen & Specker (1967) — J. Math. Mech.
- Gleason (1957) — J. Math. Mech.
- Bub & Clifton (1996) — SHPMP
- Halvorson & Clifton (1999) — IJTP
- Abramsky & Brandenburger (2011) — New J. Phys.
- McDonald & Bimbó (2023) — MLQ
- Gunji, Haruna & Uragami (2026) — arXiv:2603.22353

Topos approaches

- Isham & Butterfield (1998) — IJTP
- Döring & Isham (2008) — JMP
- Heunen, Landsman & Spitters (2009) — CMP

Delay embedding and reconstruction

- Takens (1981) — Lecture Notes in Math.
- Sauer, Yorke & Casdagli (1991) — J. Stat. Phys.
- Kennel, Brown & Abarbanel (1992) — Phys. Rev. A
- Fraser & Swinney (1986) — Phys. Rev. A
- Ragwitz & Kantz (2002) — Phys. Rev. E
- Botvinick-Greenhouse et al. (2025) — J. Stat. Phys.
- Kantz & Schreiber, *Nonlinear Time Series Analysis* (2004)

Ergodic theory

- Rokhlin (1949/1962) — disintegration
- Rokhlin (1967) — entropy lectures, Rokhlin distance
- Sinai (1959) — generating partitions
- Krieger (1970) — generator theorem
- Furstenberg (1977) — relative mixing, Szemerédi
- Shields, *Ergodic Theory of Discrete Sample Paths*

Philosophy of science

- van Fraassen, *The Scientific Image* (1980)
- van Fraassen, *Quantum Mechanics: An Empiricist View* (1991)
- de Finetti, *Theory of Probability* (1974)
- Pitowsky, *Quantum Probability — Quantum Logic* (1989)

Nonparametric statistics

- Stone (1982) — minimax rates
- Lepski (1991) — adaptive estimation
- Hoeffding (1948) — U-statistics
- Billings & Voon (1986) — NARMAX residual validation
- Casdagli (1991/1992) — deterministic vs stochastic

Model theory and probability logic

- Łoś (1955) — ultraproduct theorem
- Hoover (1978) — probability logic
- Fagin, Halpern & Megiddo (1990) — probability logic